



AQ10.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Let $f(x, y)$ be defined in a neighbourhood of (a, b) . When is (a, b) a critical point of f ?

☐

$f(a, b) = 0$

☐

$f_x(a, b) = 0$

☐

$f_x(a, b) = 0 = f_y(a, b)$

☐

f_x does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x(a, b) = 0 = f_y(a, b)$

f_x does not exist.

1 point

Consider $f(x, y) = \sqrt{5x^2 + 3y + 2}$

☐
☐

Any point on the line $x + y = 1$ in the domain of definition of f satisfies $f_y = 0$.

☐

There is no (x, y) such that $f_y = 0$.

☐

Solutions of $5x^2 + 3y + 2 = 0$ are critical points of f .

No, the answer is incorrect.

Score: 0

Accepted Answers:

There is no (x, y) such that $f_y = 0$.

Solutions of $5x^2 + 3y + 2 = 0$ are critical points of f .

1 point

A critical point

☐

is a point of maximum.

☐

may be a saddle point.

☐

may be a point of local minimum.

☐

can be obtained by equating f_x and f_y to 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

may be a saddle point.

may be a point of local minimum.



can be obtained by equating f_x and f_y to 0.

1 point

Let $f(x, y) = x^2 y$. Then

☐

$(0, 0)$ is a critical point.

☐

$(0, y)$ for $y \in \mathbb{R}$ are critical points.

☐

$f_x > 0$ at the critical points.

☐

$f_y = 0$ at the critical points.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$ is a critical point.

$(0, y)$ for $y \in \mathbb{R}$ are critical points.

$f_y = 0$ at the critical points.

1 point

For $f(x, y) = x^3 + 2xy - 2x - 4y$, which of the following is/are true?

☐

$(2, -5)$ is a critical point.

☐

$(2, 5)$ is a critical point.

☐

$f_y = 0$ is a straight line parallel to the y -axis.

☐

The system of equations $f_x = 0$, $f_y = 0$ has a unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(2, -5)$ is a critical point.

$f_y = 0$ is a straight line parallel to the y -axis.

The system of equations $f_x = 0$, $f_y = 0$ has a unique solution.

1 point

Choose the correct statement(s) about the function $f(x, y) = xy^2 + 2x$.

☐

$f_x = 0$ only for 2 values of x .

☐

$f_y = 0$ along both x and y axes.

☐

There are no critical points of f .

☐

$(0, 0)$ is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_y = 0$ along both x and y axes.

There are no critical points of f .

1 point

For $f(x, y) = \frac{x^3}{3} + y^2 + 2xy - 6x - 3y + 4$ $f(x, y) = 3x^3 + y^2 + 2xy - 6x - 3y + 4$

☐

$f_y = 0$ $f_y = 0$ is a straight line passing through the origin.

☐

$(-1, \frac{5}{2})$ $(-1, 25)$ and $(3, -\frac{3}{2})$ $(3, -23)$ are critical points.

☐

$f_x(-1, \frac{5}{2}) = 0$ $f_x(-1, 25) = 0$.

☐

$f_y(3, -\frac{3}{2}) = 2$ $f_y(3, -23) = 2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-1, \frac{5}{2})$ $(-1, 25)$ and $(3, -\frac{3}{2})$ $(3, -23)$ are critical points.

$f_x(-1, \frac{5}{2}) = 0$ $f_x(-1, 25) = 0$.

1 point

Choose the correct statement(s) about $f(x, y) = x^4 + y^4$ $f(x, y) = x^4 + y^4$.

☐

There are no critical points.

☐

There is exactly one critical point.

☐

$(0, 0)$ $(0, 0)$ is a point of global minimum.

☐

$(0, 0)$ $(0, 0)$ is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There is exactly one critical point.

$(0, 0)$ $(0, 0)$ is a point of global minimum.

$(0, 0)$ $(0, 0)$ is a point of local minimum.

Level 2

1 point

Let $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$ $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$. Then

☐

$(3, 1)$ $(3, 1)$ is a critical point.

☐

$(2, 1)$ $(2, 1)$ is a critical point.

☐

Value of f at $(2, 1)$ $(2, 1)$ is -4.

☐

Value of f at $(3, 1)$ $(3, 1)$ is -9.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(3, 1)$ $(3, 1)$ is a critical point.

Value of f at $(2, 1)$ $(2, 1)$ is -4.

1 point

Let $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$ $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$ be defined on the box $0 \leq x \leq 20$ $0 \leq x \leq 2$ and $0 \leq y \leq 10$ $0 \leq y \leq 1$. Then which of the following options is (are) true?

☐

Along the line joining $(0, 0)$ and $(2, 0)$, the minimum is attained at $(2, 0)$.

☐

$(0, 1)$ is a point of extremum along the line joining $(0, 1)$ and $(2, 1)$.

☐

$f(0, 1) = 1$

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Along the line joining $(0, 0)$ and $(2, 0)$, the minimum is attained at $(2, 0)$.

$(0, 1)$ is a point of extremum along the line joining $(0, 1)$ and $(2, 1)$.

1 point

Let $f(x, y) = \sin x \cos y$. Then

☐

$((2k+1)\frac{\pi}{2}, k\pi)$ for $k \in \mathbb{N}$ are critical points of f .

☐

$((2k+1)\frac{\pi}{2}, (2k+1)\frac{\pi}{2})$ for $k \in \mathbb{N}$ are critical points of f .

☐

$(k\pi, k\pi)$ for $k \in \mathbb{N}$ are critical points of f .

☐

$(k\pi, (2k+1)\frac{\pi}{2})$ for $k \in \mathbb{N}$ are critical points of f .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$((2k+1)\frac{\pi}{2}, k\pi)$ for $k \in \mathbb{N}$ are critical points of f .

$(k\pi, (2k+1)\frac{\pi}{2})$ for $k \in \mathbb{N}$ are critical points of f .

1 point

Let $f(x, y) = e^{-x^2 + y - \frac{y^3}{3}}$. Then

☐

$f_x(1, 0) = -\frac{2}{e}f_x(1, 0) = -e$.

☐

$f_x(0, 0) = 1$

☐

$f_y(0, -1) = 0$

☐

$f_y(1, 0) = e$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x(1, 0) = -\frac{2}{e}f_x(1, 0) = -e$.

$f_y(0, -1) = 0$

1 point

Let $f(x, y) = e^{-x^2 + y - \frac{y^3}{3}}$. Then

☐

$(0, 1)$ is a critical point.

☐

$(0, -1)$ is a critical point.

☐

$(1, 1)$ is a critical point.

☐

$(1, -1)$ is a critical point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 1)$ is a critical point.

$(0, -1)$ is a critical point.

1 point

Find three positive numbers whose sum is 12 such that the sum of their squares is minimum. The mathematical expression that best defines this problem is

☐

minimize $x^2 + y^2 + z^2$ such that $x + y + z = 12$, $x, y, z > 0$.

☐

minimize $x + y + z$ such that $x^2 + y^2 + z^2 = 12$, $x, y, z \geq 0$.

☐

minimize $x^2 + y^2 + z^2$ such that $x + y + z = 12$, $x, y, z \geq 0$.

☐

minimize $x + y + z$ such that $x^2 + y^2 + z^2 = 12$, $x, y, z \geq 0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

minimize $x^2 + y^2 + z^2$ such that $x + y + z = 12$, $x, y, z > 0$.

[Check Answers](#)

Your score is: 0/14